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| <b>Caribbean Coral Reef Institute (CCRI) Water Quality Monitoring Lab</b><br><b>University of Puerto Rico – Mayagüez Campus</b><br><b>Department of Marine Sciences</b> |                 |            |  |
| Document Title                                                                                                                                                          | Document Number | Version    |                                                                                     |
| <b>SOP Dissolved Inorganic Carbon</b>                                                                                                                                   | <b>WQP 016</b>  | <b>1.6</b> |                                                                                     |

| PREPARED BY                    |                           |                                                                                    |         |
|--------------------------------|---------------------------|------------------------------------------------------------------------------------|---------|
| Agency                         | Responsible Person        | Signature                                                                          | Date    |
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| VERSION HISTORY |          |                                                                                                                                                                                                                                                  |             |
|-----------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Version         | Date     | Description                                                                                                                                                                                                                                      | Approved By |
| 1.0             | 5/10/23  | Initial Release                                                                                                                                                                                                                                  | TAC         |
| 1.1             | 8/30/23  | Included correction for $\mu\text{M}$ to $\mu\text{mol/kg}$ in SOP, changed sample bottle to 150 mL Pyrex sample bottle, and corrected grammatical errors.                                                                                       | TAC         |
| 1.2             | 10/2/23  | Changed SOP so that all samples are analyzed in $\mu\text{M}$ and converted to $\mu\text{mol/kg}$ in post analysis and updated bottle cleaning procedure for screw-top bottles.                                                                  | TAC         |
| 1.3             | 10/4/23  | Corrected $\mu\text{mol/kg}$ to $\mu\text{M}$ conversion window text.                                                                                                                                                                            | TAC         |
| 1.4             | 11/22/23 | Added instructions to label CRM after opening and minor edits to improve clarity.                                                                                                                                                                | TAC         |
| 1.5             | 6/10/24  | Revised Apollo DIC analyzer settings for faster analysis and methods for more frequent internal calibrations to reduce post-processing of data.                                                                                                  | TAC         |
| 1.6             | 9/16/24  | Included explicit analytical settings in the SOP and revised sample analyses to run 10 samples per batch with the CRM used for the first and last sample of each batch to quantify analytical precision and accuracy following each calibration. | TAC         |

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## 1. Title

Standard Operating Procedure for Dissolved Inorganic Carbon in Seawater

## 2. Applicability

This SOP applies to the determination of seawater Dissolved Inorganic Carbon (DIC) at shallow coral reef sites around Puerto Rico.

## 3. Purpose

- 3.1. Provide laboratory-specific instructions for the CCRI Water Quality Monitoring Lab personnel to analyze coral reef seawater surface and bottom seawater for the determination of DIC at shallow coral reef sites around Puerto Rico.
- 3.2. Provide water quality data to be uploaded to the EPA WQX database for use by state and federal agencies and any other parties interested in DIC data in coral reef seawater following the Quality Project Plan.

## 4. Method

Standard Method from Dickson et al. (2007).

## 5. Health and Safety

- 5.1. Follow the safety indications for equipment and personnel suggested by the manufacturer.
- 5.2. Read each reagent SDS prior to use for safe handling and disposal.

## 6. General Precautions

- 6.1. Always maintain two points of contact when deploying the Van Dorn bottle on the boat to avoid losing the bottle overboard.
- 6.2. Ensure the boat engines are idle and the captain approves of the Niskin bottle being deployed overboard before deploying the bottle to ensure that the line does not become entangled with the boat propeller.

## 7. Training

- 7.1. Personnel shall be trained in the proper sample collection, handling and analysis procedure for DIC.
- 7.2. Personnel shall receive training on how to complete the field sampling log for this SOP.

## 8. Required Materials

### 8.1. Sampling Equipment

- 8.1.1. Van Dorn Bottle
- 8.1.2. Sampling line
- 8.1.3. Van Dorn messenger
- 8.1.4. Tygon Tubing

### 8.2. Consumables

- 8.2.1. Ultrapure Nitrogen Compressed Gas Cylinder
- 8.2.2. Phosphoric Acid

|                                                                                                                                                                         |                 |            |                                                                                     |
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- 8.2.3. Sodium Chloride
- 8.2.4. Kimwipes
- 8.2.5. Parafilm
- 8.2.6. Distilled Water
- 8.2.7. Sodium Bicarbonate
- 8.2.8. Certified Reference Material for Oceanic CO<sub>2</sub> measurements
- 8.2.9. Glass Storage Bottle (25 mL)
- 8.2.10. Glass Storage Bottle (500mL)
- 8.2.11. Glass Storage Bottle (1000mL)
- 8.2.12. 150 mL Pyrex Screw-top Bottles
- 8.2.13. Mercuric (II) Chloride
- 8.2.14. Pipette (10-100µL)
- 8.2.15. Pipette Tips (100µL)
- 8.2.16. Traceable High Accuracy Thermometer

### 8.3. Dissolved Inorganic Carbon Analyzer System

- 8.3.1. Apollo Scitech AS-C5 Dissolved Inorganic Carbon Analyzer
- 8.3.2. DIC AS-C5 software

### 8.4. Chemicals & Solutions Preparation

| Compound             | CAS No.   |                                          | Safety classification                                                                                                        |
|----------------------|-----------|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Mercuric chloride    | 7487-94-7 | <i>GHS03, GHS05, GHS06, GHS07, GHS09</i> | Danger - Oxidizing, Warning/Danger-Corrosive, Corrosive, Irritant, Danger – Toxic, Warning/Danger- Environmentally Damaging. |
| Phosphoric Acid, 85% | 7664-38-2 | <i>GHS05, GHS07</i>                      | Warning/Danger – Corrosive, Irritant                                                                                         |
| Sodium Chloride      | 1310-72-2 | <i>GHS05</i>                             | Warning/Danger                                                                                                               |
| Sodium Bicarbonate   | 144-55-8  | –                                        | Not a hazardous substance                                                                                                    |

#### 8.4.1. Nitrogen Carrier Gas

| Reagent                | Amount                                                                                                                           | Preparation Instructions                                                                                                                        |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Nitrogen gas           | Cylinder                                                                                                                         | The nitrogen should be of ≥99.998% purity.<br>Put two-stage regulator capable of delivering a stable pressure of 16 – 17 psi (1 atm = 14.5 psi) |
| <b>General Details</b> |                                                                                                                                  |                                                                                                                                                 |
| Storage                | Store in a well ventilated place attached to the wall with a secure chain and connected to the DIC analyzer with a proper tubing |                                                                                                                                                 |
| Stability              | When finish replace for a new one                                                                                                |                                                                                                                                                 |

|                                                                                                                                                                         |                 |            |                                                                                     |
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|               |         |
|---------------|---------|
| Reagent Label | Warning |
|---------------|---------|

#### 8.4.2 Phosphoric acid (3%) with NaCl (7%)

| Reagent                   | Amount  | Preparation Instructions                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------------------------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 85% Phosphoric acid       | 35.6 mL | <ol style="list-style-type: none"> <li>1. Weigh 70 g of NaCl, transfer in a beaker of 1L.</li> <li>2. Measure 600 mL DI water and add to the beaker containing the NaCl.</li> <li>3. Measure 35.6 mL of phosphoric acid in a 50 mL cylinder and transfer it into the beaker containing the NaCl and the DI water.</li> <li>4. Mix well with a stirring bar.</li> <li>5. Transfer the mix in a cylinder of 1 L. Bring the volume to 1 L with DI water</li> </ol> |
| 7% Sodium Chloride (NaCl) | 70 g    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| DI Water                  | 1 L     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

| General Details |                     |
|-----------------|---------------------|
| Storage         | At room temperature |
| Stability       | Indefinitely        |
| Reagent Label   | Warning             |

#### 8.4.3 Mercuric (II) chloride saturated solution

| Reagent               | Amount                                                         | Preparation Instructions                                                                                                                                                                                                                                                    |
|-----------------------|----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mercury (II) Chloride | 2.0 g                                                          | <ol style="list-style-type: none"><li>1. Add the 2 g HgCl<sub>2</sub> and 25 mL H<sub>2</sub>O to a 25 mL Pyrex bottle.</li><li>2. Close bottle, shake to mix thoroughly, and label “Danger HgCl<sub>2</sub> solution”.</li><li>3. Place in a sealed plastic bag.</li></ol> |
| Distilled water       | 25.0 mL                                                        |                                                                                                                                                                                                                                                                             |
| General Details       |                                                                |                                                                                                                                                                                                                                                                             |
| Storage               | 25 mL glass bottle at room temperature in a sealed plastic bag |                                                                                                                                                                                                                                                                             |
| Stability             | Indefinitely                                                   |                                                                                                                                                                                                                                                                             |
| Reagent Label         | Warning, Danger, Toxic                                         |                                                                                                                                                                                                                                                                             |

|                                                                                                                                                                         |                 |            |                                                                                     |
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#### 8.4.4 Sodium bicarbonate solution (2000 µM)

| Reagent            | Amount                                  | Preparation Instructions                                                                                                                                                                                                                                                                                                                        |
|--------------------|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sodium bicarbonate | 0.084 g                                 | <ol style="list-style-type: none"><li>1. Weigh 0.084 g of sodium bicarbonate.</li><li>2. Measure 500 mL distilled water in a graduated cylinder.</li><li>3. Add the 0.084 g of sodium bicarbonate and the 500 mL distilled water to a 500 mL Pyrex bottle.</li><li>4. Mix thoroughly and label “2000 μM sodium bicarbonate solution”.</li></ol> |
| Distilled water    | 500 mL                                  |                                                                                                                                                                                                                                                                                                                                                 |
| General Details    |                                         |                                                                                                                                                                                                                                                                                                                                                 |
| Storage            | 500 mL glass bottle at room temperature |                                                                                                                                                                                                                                                                                                                                                 |
| Stability          | Indefinitely                            |                                                                                                                                                                                                                                                                                                                                                 |
| Reagent Label      | N/A                                     |                                                                                                                                                                                                                                                                                                                                                 |

## 9. Procedure

### 9.1. Sample Collection

- 9.1.1. Prepare a Van Dorn bottle to be deployed.
- 9.1.2. Lower the Van Dorn bottle in a slow and steady mode until it gets close to the bottom with caution not to touch the bottom and stir sediment up.
- 9.1.3. Drop the bottle messenger to close the Van Dorn bottle at the desired depth.
- 9.1.4. Pull the Van Dorn bottle back up into the boat.
- 9.1.5. Connect the tubing to the Van Dorn bottle spigot to transfer water.
- 9.1.6. Push open the Van Dorn bottle's spigot valve to allow water flow.
- 9.1.7. Place the Tygon tubing at the bottom of the inverted 150 mL screw-top Pyrex glass bottle to rinse with water from the Van Dorn bottle.
- 9.1.8. Flip the sample bottle opening side up to begin filling the sample bottle from the bottom.
- 9.1.9. Collect the sample until the bottle overflows 1.5 times the sample volume.
- 9.1.10. Pinch the Tygon tubing to stop flow and remove the tubing from the sample vial. This will leave approximately 1% head space in the bottle.
  - 9.1.10.1. Refer to Appendix J for the seawater sample collection order.
- 9.1.11. Poison the sample.
  - 9.1.11.1. Add 50 µL mercury (II) chloride (HgCl<sub>2</sub>) to the 150 mL sample bottles.
- 9.1.12. Close the bottle.
  - 9.1.12.1. Ensure that the sample is filled to the top of the bottle and screw the lid on tightly to prevent any gas exchange.

|                                                                                                                                                                         |                 |            |                                                                                     |
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- 9.1.13. Gently invert the bottle several times to mix the mercuric chloride with the sample.
- 9.1.14. Place the sample bottle inside the cooler until analysis in the lab.

## 9.2. Instrument Start-up

- 9.2.1. Ensure that the LICOR is on.
- 9.2.1.1. The circular silver switch (LICOR) on the front of the instrument is depressed with a blue light when on.
- 9.2.1.2. Turn on LICOR if needed by pressing the button.
- 9.2.1.3. The LICOR should be allowed to warm up overnight or longer for best results and ideally left on continuously.
- 9.2.2. Turn the main system on (“MAIN” switch on front of instrument below the silver LICOR switch).
- 9.2.3. Open the N<sub>2</sub> carrier gas using the knob on the top of the tank and ensure the pressure of the gas is between 16-17 psi.
- 9.2.3.1. Use the main gas knob on the N<sub>2</sub> tank to establish gas flow.
- 9.2.3.2. Open the main gas knob fully and then close ¼ turn for safety.
- 9.2.3.3. Do not adjust the other knobs unless the gas pressure is not between 16-17 psi.
- 9.2.4. Turn on the DIC analyzer computer and open DIC ASC5 software.
- 9.2.5. Click the “Initialize” button once to start communications between the software, syringe pump, and LICOR.
- 9.2.5.1. Use the pop-up box to enter the file name for the day’s analysis
- 9.2.6. Wait until the temperature display on the front of the instrument is 2-3°C.
- 9.2.7. Ensure the CO<sub>2</sub> (ppm) is correctly displaying data.
- 9.2.8. Go to the general setting tab on the left panel. Check that the system displays “Ready” in the top right panel ( in the text box above the CO<sub>2</sub> vs time plot). Use the Analytical Settings “**Analytical\_Settings\_Fast**” and do not modify any of the values.

- 9.2.8.1. Analytical\_Settings\_Fast uses the following settings:

|                        |             |
|------------------------|-------------|
| Sample volume (ml)     | <b>1.00</b> |
| Wait Btwn Repeat (s)   | <b>10</b>   |
| Wait Btwn Sample (s)   | <b>5</b>    |
| Flush Volume (ml)      | <b>0.70</b> |
| Pre-Acid Volume (ml)   | <b>0.70</b> |
| Acid Volume (ml)       | <b>0.70</b> |
| Injection Speed (sps)  | <b>2000</b> |
| End Analysis - BL Diff | <b>0.01</b> |

|                                                                                                                                                                         |                 |            |                                                                                     |
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|                            |                               |
|----------------------------|-------------------------------|
| End Analysis - Deviation   | <b>0.2</b>                    |
| Precision <b>0.001 + 0</b> | Best <b>2</b> Out of <b>5</b> |
| Standard Volumes (ml)      | <b>0.80   1.00   1.20</b>     |
|                            | <b>Linear Fit</b>             |

9.2.9. Gently swirl the acid solution and internal standard (2000  $\mu\text{M}$   $\text{NaHCO}_3$ ) bottles, remove the caps, and place the rubber stoppers with holes for tubes.

9.2.9.1. Check to ensure that there are no particulates or growth in the acid solution or internal standard bottles before proceeding as these can lead to poor precision.

9.2.10. Place all standard and sample tubes into the internal standard sample solution (2000  $\mu\text{M}$   $\text{NaHCO}_3$ ) and the acid tube into DI water.

9.2.11. Click the Clean Valve (x1) button once to prime all sample tubes with internal standard sample solution (2000  $\mu\text{M}$   $\text{NaHCO}_3$ ) and acid tube with DI water and wait until complete.

9.2.11.1. Click the x1 or x2 next to the button to toggle between x1 or x2 options.

9.2.11.2. The top left status bar above the LICOR  $\text{CO}_2$  plot will display "Ready" on the right side of the panel in the Plot window when completed.

9.2.12. Leave the standard intake tube (Port C, blue knob) in the internal standard sample solution (2000  $\mu\text{M}$   $\text{NaHCO}_3$ ) so that the tube is near the bottom of the bottle.

9.2.13. Place the acid intake tube (Port L, thicker tubing) into the 3% phosphoric acid + 7% sodium chloride solution so that the tube is near the bottom of the bottle.

9.2.14. Proceed to Section 9.3 Sample Analysis.

### 9.3. **Sample Analysis**

9.3.1. Ensure that all samples, CRM, and reagents are at room temperature and record the temperature of the lab using a thermometer.

9.3.2. Click the "Single" button on the left window panel.

9.3.3. Set the Single Analysis Volume to 1.0 mL in the Single Analysis Port "C (STD)"

9.3.4. Leave the standard line C the internal standard sample solution (2000  $\mu\text{M}$   $\text{NaHCO}_3$ ) solution.

9.3.5. Click "Load" to load the sample for analysis.

9.3.6. Click "Start Analysis" to begin analyzing samples.

9.3.7. Repeat 3 times to ensure that precision of 2 replicate samples is within the desired  $\pm 3 \mu\text{M}$ .

9.3.7.1. If the desired precision target is met, proceed to step 9.3.7. Otherwise, run two additional samples to see if precision is within  $\pm 3 \mu\text{M}$ .

|                                                                                                                                                                         |                 |            |                                                                                     |
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- 9.3.7.2. If two replicate samples out of the five total samples are not within  $\pm 3$   $\mu\text{M}$ , see instrument troubleshooting manual.
- 9.3.8. Click the “Batch” button on the left window panel.
- 9.3.9. Set the “# of Samples in Batch” = 10.
  - 9.3.9.1. This will run 10 samples between every calibration curve.
- 9.3.10. Change the CRM DIC ( $\mu\text{M}$ ) to the respective concentration for the CRM used.
  - 9.3.10.1. The CRM DIC values for the respective CRM Batch numbers can be accessed via:  
[https://www.ncei.noaa.gov/access/ocean-carbon-acidification-data-system/oceans/Dickson\\_CRM/batches.html](https://www.ncei.noaa.gov/access/ocean-carbon-acidification-data-system/oceans/Dickson_CRM/batches.html)
  - 9.3.10.2. **The CRM DIC values are reported in  $\mu\text{mol/kg}$  on the NOAA website and must be converted to  $\mu\text{M}$  before entering the value into the Apollo system software.**
  - 9.3.10.3. Press the circling arrows to access the DIC unit conversion window.
  - 9.3.10.4. Input the reported DIC of the CRM in  $\mu\text{mol/kg}$ .
  - 9.3.10.5. Input the Salinity of the CRM.
  - 9.3.10.6. Click “OK” to accept the calculated CRM in units of  $\mu\text{M}$ .
- 9.3.11. Open a new CRM bottle, wrap the standard port “C” line in a KimWipe to prevent it from getting grease on the line, and place the line into the bottom of the CRM bottle.
  - 9.3.11.1. Label the CRM bottle with the date opened and your initials.
- 9.3.12. Samples should only be open directly before analysis to avoid compromising the samples to gas exchange.
- 9.3.13. Click “Load” to load the 10 samples for analysis.
- 9.3.14. Change the “Sample Name” to the correct sample name of the respective seawater sample.
  - 9.3.14.1. The CRM sample name should match the bottom of the CRM used for analysis. The bottle number is next to the barcode in the following format “CRM\_BatchNumber\_BottleNumber”
- 9.3.15. Place the corresponding sample tubing analysis into the respective sample bottle.
- 9.3.16. Check that the bottle name matches the sample name for the respective Tecan port for each sample (D, E, F, G, H, I, J, K).
  - 9.3.16.1. For each sample, open the sample bottle, cover the opening with Parafilm, poke a hole in the Parafilm, and place the line through the hole and close to the bottom of the bottle.
  - 9.3.16.2. Samples should only be open directly before analysis to avoid compromising the samples to gas exchange.
  - 9.3.16.3. A CRM should be analyzed as the first sample immediately after each calibration to ensure that the system is precise and accurate following the initial calibration and then at the end of each calibration to verify the precision and accuracy of each set of samples.

|                                                                                                                                                                         |                 |            |                                                                                     |
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- 9.3.16.3.1. Set the port to C (STD) and replace the sample name with the CRM bottle number.
- 9.3.16.4. The last sample analyzed on each day should be a CRM to ensure accuracy of all samples.
- 9.3.17. Click Start Analysis.
- 9.3.18. Ideally all samples should be analyzed at room temperature to minimize differences in density between samples.
  - 9.3.18.1. Record the temperature of the room using the traceable handheld thermometer.
  - 9.3.18.2. Record the temperature of each sample after DIC analysis and record the Date, Time, Sample ID, Temperature, and DIC ( $\mu\text{M}$ ) in the DIC laboratory notebook.
    - 9.3.18.2.1. Note, the CRM is stored at room temperature and therefore the CRM temperature is assumed to be the same as the room.
    - 9.3.18.2.2. The samples are preserved in the refrigerator and therefore sample temperature must be recorded for those samples.
- 9.3.19. Occasionally, a sample analysis may fail the precision quality assurance protocols for replicate measurements established by the analysis settings resulting in an error message.
  - 9.3.19.1. "There was a failed sample analysis. Please ensure the Instrument and samples are configured correctly."
  - 9.3.19.2. Click "OK"
  - 9.3.19.3. The instrument will repeat the analysis of the failed sample.
  - 9.3.19.4. If samples repeatedly fail, the filters and tubing on the back of the DIC analyzer likely need to be replaced and cleaned to remove any salt build-up.
- 9.3.20. Repeat steps 9.3.13 to 9.3.19 for any additional samples as needed.
- 9.3.21. The last sample analyzed on each day should be a CRM to ensure accuracy of all samples.
- 9.3.22. Click "Delete All Pending" after the final CRM is analyzed to end the analyses for the day.
- 9.3.23. The instrument will display a message when complete saying "There are no pending samples."
- 9.3.24. Click OK.
- 9.4. Instrument Powering Off**
  - 9.4.1.** Place all tubes in DI water.
  - 9.4.2.** Go to General Settings.
  - 9.4.3.** Click the "Clean Valve x2" button once and wait until complete.
    - 9.4.3.1. For each step, the system will say "Ready" at the top left corner when the operation is completed.
  - 9.4.4.** Toggle the Clean Valve button from x2 to x1.

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- 9.4.5. Click the “Clean Valve x1” button once and wait until complete.
- 9.4.6. Click the “Purge” button once and wait until complete.
- 9.4.7. Remove all tubes from the DI water.
- 9.4.8. Click the “Purge” button once and wait until complete.
- 9.4.9. Click the “Clean Valve x2” button once and wait until complete.
- 9.4.10. Click the “X” button at the top of the screen.
- 9.4.11. A confirmation message will ask you to confirm that the valve has been cleaned using the purge and clean valve functions, click OK if you have already completed those steps to exit the program.
- 9.4.12. Turn off the N<sub>2</sub> carrier gas using the gas knob on the top of the regulator.
- 9.4.13. Turn off the main system using the “MAIN” toggle switch on the front of the DIC Analyzer.
- 9.4.14. Leave the LICOR on (circle switch with blue light) to keep the system warm for best results when next analyzing samples.

## 10. Documentation

- 10.1. Field data will be recorded in the Sampling Log (Appendix F).
- 10.2. Field Sampling Log (Appendix F) shall be kept in the designated file cabinet at CCRI office.
- 10.3. Data contained in hard copy form shall be digitized into the corresponding electronic spreadsheets as soon as possible after the task is completed.
- 10.4. All spreadsheets should be uploaded to the internal CCRI Water Quality Database.
- 10.5. All data will be electronically archived and backed up online.

## 11. Cleaning and Maintenance

- 11.1. If the analyzer will be left unused for an extended period of time, it is recommended to clean the 12-way valve with DI water and air weekly.
- 11.2. Bottle cleanup process:
  - 11.2.1. Submerge bottles and stoppers in a solution of Liquinox and water for several hours.
    - 11.2.1.1. Dilute Liquinox 1:100 on water.
  - 11.2.2. Remove bottles from the Liquinox solution and rinse with tap water three times.
  - 11.2.3. Rinse the interior of the bottle with distilled water three times.
  - 11.2.4. Place the bottles and stoppers on a drying rack to completely air dry.
    - 11.2.4.1. Place the bottles in an inverted position to remove water from the interior.
  - 11.2.5. Place the caps in a position to air dry.

## 12. Troubleshooting

- 12.1. See instrument manual for troubleshooting and contact the manufacturer Apollo Scientific ([admin@apolloscitech.com](mailto:admin@apolloscitech.com)) if the problem persists.

## 13. QA/QC

- 13.1. As per QAPP Table 4, the DIC from seawater should be measured in µmol/kg with a precision of <20%.

|                                                                                                                                                                         |                 |            |                                                                                     |
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13.1.1. The instrument measures samples volumetrically in units of  $\mu\text{M}$  so all samples must be converted to  $\mu\text{mol/kg}$  for the final report:

13.1.1.1.  $\text{DIC } (\mu\text{mol/kg}) = \text{DIC } (\mu\text{M}) / \text{density}_{\text{sample}} (\text{kg/L})$

13.1.1.2. Sample density can be calculated following the equation by Millero & Poisson (1981) as expressed in the Apollo Dissolved Inorganic Carbon Analyzer Instruction Manual.

13.2. Any seemingly erroneous measurements will be flagged for further review in accordance with the QAPP.

## 14. References

- 14.1. Dickson AG, Sabine CL, Christian JR. Guide to best practices for ocean  $\text{CO}_2$  measurements. North Pacific Marine Science Organization; 2007.
- 14.2. Apollo SciTech LLC. Model AS-C5 Dissolved Inorganic Carbon Analyzer Instruction Manual 2022.11